Enhancing Contrast in Circular-view Photoacoustic Computed Tomography Systems

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***Abstract*— In this study, we propose a novel method to enhance image quality in sparse circular-view photoacoustic computed tomography (PACT) systems. Traditional PACT systems rely on dense arrays of detectors to achieve high-resolution images, but they suffer from high cost and computational complexity. Our proposed method addresses these challenges by reconstructing multiple low-quality images (LQIs) from under-sampled data and using advanced signal processing techniques, including cross-correlation and the coefficient of variation (CV), to generate a stability ratio (SR) that significantly improves image quality and contrast. We evaluated the proposed method using both artery and Derenzo phantoms, with detector arrays of 1024, 512, and 256 elements. The results demonstrated that our approach effectively suppresses artifacts and enhances image quality even with a reduced number of detectors. Quantitatively, the proposed method showed substantial improvements over conventional techniques, with up to 50% reduction in background noise, a 6 dB increase in signal-to-noise ratio (SNR), and a significant enhancement in structural similarity index (SSIM). These findings suggest that the proposed method enables high-quality PACT imaging with fewer detectors, which is essential to develop a cost-effective and efficient biomedical imaging tool.**

Keywords—Circular-View PACT, Sparse, Photoacoustic Imaging, Image Reconstruction, Artifact Suppression.

I. Introduction

Photoacoustic (PA) computed tomography (PACT) is an advanced imaging modality that integrates the high contrast of optical imaging with the deep penetration and spatial resolution of ultrasound imaging [1, 2]. A specialized form, the circular-view PACT system, employs a circular array of ultrasonic transducers surrounding the target tissue [3]. This circular configuration is widely utilized in various medical applications, including breast imaging, cardiovascular studies, and neuroscience, due to its capability for high-resolution, non-invasive visualization of functional and molecular information of tissues and organs [4, 5].

In circular-view PACT systems, illumination of the target tissue is typically achieved using short-pulsed lasers [6]. The laser energy is absorbed by chromophores such as hemoglobin, melanin, or exogenous contrast agents, leading to localized thermoelastic expansion and the generation of ultrasound waves [7]. A circular array of transducers detects these acoustic waves from all directions, enabling comprehensive imaging and high-resolution tomographic reconstruction [8, 9]. The circular array geometry of detectors enhances spatial resolution and ensures isotropic imaging, crucial for accurate diagnostics [3, 10]. Multi-angle detection captures acoustic signals with high fidelity, which is essential for precise image reconstruction [11, 12].

However, the complexity and cost of circular-view PACT systems, especially with high element count, pose practical challenges. The requirement for a high density of detectors to ensure isotropic imaging results in increased system size, complexity, and higher costs for manufacturing and maintenance [8, 13]. Precise synchronization and calibration of numerous detectors add to the complexity, potentially leading to errors and artifacts if not managed properly [14, 15]. These challenges limit the adoption of circular-view PACT systems in clinical settings.

Sparse circular-view PACT systems have emerged as a solution to these challenges. By strategically positioning fewer detectors around the circular path, these systems reduce complexity and costs while maintaining high-quality image reconstruction [16]. Advanced reconstruction algorithms, including compressed sensing, beamforming, and machine learning, are employed to reconstruct detailed images from limited data [17]. Recent studies have been focused on optimizing sparse detector configurations to balance detector reduction with image quality enhancement [18]. Techniques such as coherent compounding and optimized detector placement help lower system complexity and costs [19, 20]. Signal processing methods like time-reversal and model-based reconstruction further improve contrast by refining and localizing photoacoustic signals [17, 21-23]. Sparse reconstruction techniques, effective in low signal-to-noise ratio conditions, enhance the visualization of subtle features, while noise reduction strategies, including wavelet filtering, machine learning [24-27], and weighting factors [28-30] minimize background noise and highlight critical details.

In this study, we propose a novel method to enhance image stability and contrast in sparse circular-view PACT systems. Our approach involves reconstructing multiple low-quality images (LQIs) from under-sampled acquired data, calculating cross-correlations to assess consistency, and computing the coefficient of variation (CV) for pixel values. This CV is inverted and normalized to produce a stability ratio (SR), which is then applied to the initial reconstructed image to improve stability and overall quality.

II. Materials and Methods

In circular-view PACT systems, determining the minimum number of detection points is crucial for accurate image reconstruction and adherence to the Nyquist criterion, which states that the sampling rate must be at least twice the highest frequency component in the signal to avoid aliasing [31]. Therefore, for a typical circular-view PACT imaging system using a ring array transducer with a radius of 5 cm, an upper frequency of 3 MHz, and a sound speed of 1500 m/s, the minimum number of detection points required to meet the Nyquist criterion is approximately 1250.

In PACT, when optical absorbers are excited, high initial pressure are generated within the region that is excited, and generated PA waves propagate omnidirectionally. The ring array transducer captures these PA waves, which are then back-projected as circular arcs in the ROI using the back-projection reconstruction method for every detector position. Consequently, back-projecting the recorded PA signals from each acoustic transducer (AT) results in arc-shaped artifacts corresponding to the high-intensity PA absorbers. PACT reconstruction with fewer ATs, results in low-quality-image (LQI). Each LQI is reconstructed independently using different ATs. Therefore, streak artifacts are expected to vary in position across different LQIs. By calculating the cross-correlation of LQIs and then determining the CV of these cross-correlation values, we anticipate that the primary signals from PA absorbers will be enhanced, while streak artifacts will be effectively suppressed.

The detailed workflow of the proposed method is outlined in the following steps:

Step 1: Reconstruct the initial image using data from all the available ATs. We begin by applying a conventional delay-and-sum algorithm to the data from all detectors to produce the initial image [32, 33].

Step 2: Reconstruct PA images using sampled data from scattered ATs. We generate a minimum of three PA images from the undersampled data employing a conventional delay-sum algorithm, as at least three images are required to calculate the CV.

Step 3: Calculate pixel-wise cross-correlations for each pair of reconstructed images. We compute cross-correlations to assess the consistency and quality of the reconstructed images.

Step 4: Compute the CV for all pixels within the imaging target based on the cross-correlation results. This step quantifies the variability in pixel values across the different reconstructed images. The CV formula for pixel (x, y) in the imaging domain is as follows:

$$\mathrm{CV}_{(x,y)} = \frac{\sigma_{\text{cross-corr}}}{\mu_{\text{cross-corr}}}, \tag{1}$$

where, σ is the standard deviation of the output values from the cross-correlation of the images, and μ is the mean of the output values from the cross-correlation of the images.

Step 5: Invert the CV and normalize the data to a range of [0, 1], resulting in the SR. This normalization process transforms the variability measure into a stability metric that highlights consistent areas.

Step 6: Apply the SR to the reconstructed image by all ATs. The SR is used to adjust the initial image, enhancing the visibility of stable features while suppressing artifacts, thereby improving overall image quality.

This approach enables effective image reconstruction and contrast enhancement by leveraging under-sampled data and advanced image processing techniques.

III. Numerical Studeis and Results

A. Numerical study

The raw PA signals were generated using the k-Wave MATLAB toolbox [34]. To evaluate the proposed method, we synthesized images of the Derenzo and Artery phantoms. To compare and assay the performance of the proposed method, three different number of detectors (1024, 512, and 256) were considered. The synthesized ring array had a curvature radius of 50 mm, to capture PA signals at a sampling frequency of 25 MHz. Following data acquisition, the PA signals were processed through a Gaussian band-pass filter with a center frequency of 2.25 MHz and a fractional bandwidth of 70%. The media's grid size was set to 0.01 mm in both X and Y directions, and the phantom's empty media was simulated with water, having a sound speed of 1500 m/s. For quantitative assessment, we calculated the structural similarity index (SSIM) and signal-to-noise ratio (SNR) for images reconstructed using both conventional and proposed methods.

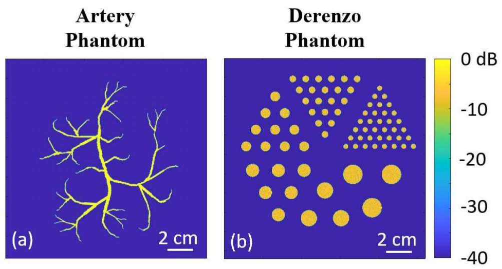

Fig. 1. (a, b) The schematic of the Artery and Derenzo phantoms, respectively.

B. Numerical Results

1) Artery phantom

Figure 2 presents the numerical results for the artery phantom. The reconstructed images using the conventional method and the proposed method with three LQIs are shown in **Figs. 2(a)** and **2(b)**, respectively. The conventional approach employed detector arrays with 1024, 512, and 256 elements. As the number of detectors decreased, noticeable artifacts emerged in the reconstructed images, particularly around the photoacoustic absorbers, as illustrated in **Figs. 2(b1-b3)**. The proposed method was evaluated under two sparse configurations, using 512 and 256 detector elements, with the resulting SR images displayed in **Fig. 2(c)**. Compared to the conventional method, the proposed approach effectively suppressed artifacts in densely populated regions and significantly enhanced overall image quality, as demonstrated in **Figs. 2(b)** and **2(c)**. **Table I** provides a quantitative comparison between the two methods, highlighting the substantial improvements achieved by the proposed method. Specifically, the standard deviation of background noise was reduced from 7.7 to 4.3 ($\times 10^3$), SNR (mean value of the signal to standard deviation of the background) increased from 25 dB to 31 dB, and SSIM improved from 83% to 92% at 1024 detection points. Even with 256 detection points, the proposed method significantly reduced background noise [19 vs. 44 ($\times 10^3$)], increased SNR (24 dB vs. 11 dB), and markedly enhanced SSIM (84% vs. 11%) compared to the conventional approach.

2) Derenzo phantom

Figure 3 shows the numerical results for the Derenzo phantom, with reconstructed images obtained using the conventional method and the proposed method with three LQIs shown in **Figs. 3(a)** and **3(b)**, respectively. Following a similar methodology, the conventional approach employed detector arrays with 1024, 512, and 256 elements. As the number of detectors was reduced, the images exhibited increasing artifacts, particularly around the photoacoustic absorbers, as illustrated in **Figs. 3(b1-b3)**. The proposed method was also tested under two sparse configurations, using 512 and 256 detector elements, with the corresponding SR images presented in **Fig 3(c)**. Similar to the artery phantom results, the proposed method demonstrated superior performance, effectively reducing artifacts in densely populated regions and improving overall image quality, as shown in **Figs. 3(b)** and **3(c)**. **Table II** provides a quantitative comparison between the conventional and proposed methods. For the Derenzo phantom, the proposed method significantly enhanced the results, reducing background noise from 13 to 9 ($\times10^3$), increasing SNR from 21 dB to 25 dB, and improving SSIM from 64% to 80% at 1024 detection points. Even with 256 detection points, the proposed method achieved better results, lowering background noise [20 vs. 48 ($\times10^3$)], increasing SNR (18 dB vs. 10 dB), and substantially improving SSIM (59% vs. 10%) compared to the conventional method.

TABLE I
THE MEASURED QUANTITATIVE VALUES

Num. of ATs	SSIM [%]		SNR [dB]		Background (std ×1e3)	
	Conv.	Prop.	Conv.	Prop.	Conv.	Prop.
1024	83	92	25	31	7.7	4.3
512	41	90	19	28	17	6
265	11	84	11	24	44	19

TABLE II
THE MEASURED QUANTITATIVE VALUES

Num. of ATs	SSIM [%]		SNR [dB]		Background (std $\times10^3$)	
	Conv.	Prop.	Conv.	Prop.	Conv.	Prop.
1024	64	80	21	25	13	9
512	28	78	16	25	22	8
265	10	59	10	18	48	20

IV. Discussion and Conclusion

The results presented in this study demonstrate the effectiveness of the proposed method in enhancing PACT image quality while using lower channel count acquisition systems. By employing a novel strategy that involves reconstructing multiple LQIs from under-sampled data, calculating cross-correlations between image pairs, and using the CV to generate a SR, we have shown significant improvements in image quality, even when the number of detection points is substantially reduced.

In the case of the artery phantom, the proposed method outperformed the conventional approach across all tested configurations, particularly in scenarios with fewer detectors. The method effectively suppressed streak artifacts and other unwanted noise that typically arise in densely populated regions, leading to enhanced image contrast and clarity. Notably, even with as few as 256 detection points, the proposed method maintained high SSIM and SNR, demonstrating its robustness against data sparsity.

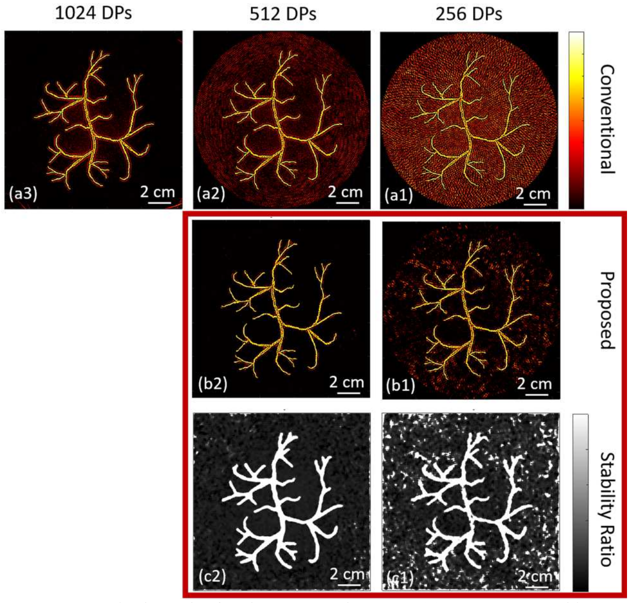

Fig. 2. Numerical results for the Artery phantom. (a) Reconstructed image using the conventional method. (b) Reconstructed image using the proposed method. (c) Obtained SR values with the proposed method

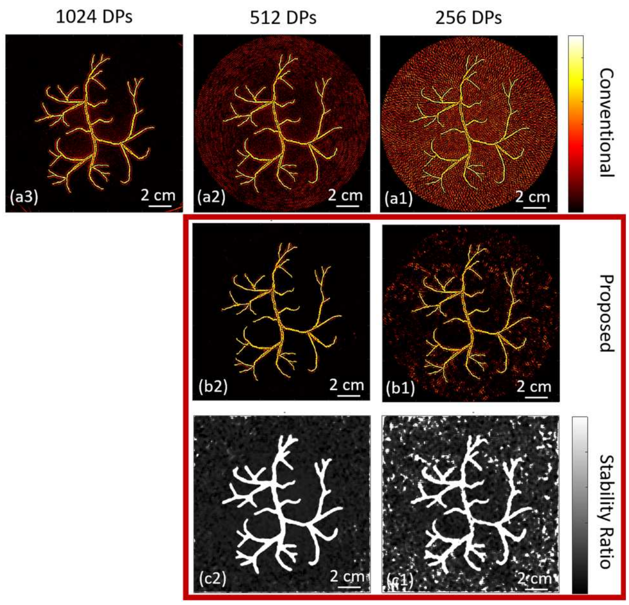

Fig. 3. Numerical results for the Derenzo phantom. (a) Reconstructed image using the conventional method. (b) Reconstructed image using the proposed method. (c) Obtained SR values with the proposed method

Similarly, the Derenzo phantom results confirmed the proposed method's superiority in sparse settings. The conventional method struggled with artifact suppression as the number of detectors decreased, leading to significant degradation in image quality. In contrast, the proposed approach maintained lower background noise and higher SSIM and SNR values, even under challenging sparse configurations. This indicates the proposed method's ability to extract and enhance the essential features of the imaging target, despite limited data availability.

By using cross-correlation and CV inversion, the proposed method enables the system to overcome the inherent limitations of sparse detection, providing reliable and accurate imaging results. The SR, which normalizes the inverted CV, serves as a powerful tool for highlighting stable regions in the reconstructed images, further improving the final image quality.

In conclusion, the proposed method offers a promising solution for enhancing image quality in sparse circular-view PACT systems. It allows for significant reductions in the number of detectors without sacrificing image fidelity, making PACT systems more cost-effective and practical for various biomedical applications. Future work could focus on further optimizing the algorithm for real-time imaging and exploring its applicability to other types of phantoms and *in-vivo* imaging scenarios.

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